



Unveiling the potential: Exploring the predictability of complex exchange rate trends

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ABSTRACT

Forecasting exchange rates is challenging due to its diverse features and complex patterns. Inspired the theory of receptive fields, we proposed two models: the Transformer with Convolutional Neural Network (CNN-Transformer) and the Long Short Term Memory networks with Convolutional Neural Network (CNN-LSTM). These models leverage Convolutional Neural Network (CNN) modules to expand the learnable timestep of Long Short Term Memory networks (LSTM) and reduce the complexity of Transformer. Generally, CNN-LSTM demonstrates the highest predictive accuracy. Furthermore, there is a silver lining: the sentiment in past news is closely linked to future exchange rates. In order to introduce the news sentiment to improve the performance of models, we fine-tune a pre-trained model, Bidirectional Encoder Representations from Transformers with Whole Word Masking (BERT-WWM), to extract the sentiment information from news dataset. However, the effectiveness of introducing news sentiment is greatly dependent on the timesteps. Longer timesteps often increase the likelihood of success for this method. We believe this pattern relates to the timeliness of news and the delayed impact of news events. Also, it's important to note that the content of news text significantly influences the forecasting performance.

1. Introduction

In the era of increased globalization and economic integration, the significance of understanding and accurately forecasting exchange rates cannot be overstated. The foreign exchange (Forex) market, as the largest financial market globally, boasts a staggering daily turnover exceeding 7.5 trillion dollars (McGuire et al., 2022). Foreign exchange serves as a vital financial instrument for international trade and economic activities, exerting a profound influence on the balanced development of domestic and foreign economies. It serves as a reflection of the relationship between a country or region and the broader international economic environment. Given its crucial role, the ability to comprehend the fundamental elements driving exchange rate determination, grasp the patterns of exchange rate fluctuations, and forecast exchange rate trends is paramount for countries, enterprises, and individual investors.

In recent years, researchers have dedicated their efforts to tackling the challenge of predicting complex trends in exchange rates. These endeavors have yielded promising results, indicating that these intricate

patterns can indeed be forecasted (Ackermann et al., 2017). Meanwhile, deep learning extended the range of prediction models, which are different from previous statistical prediction models (e.g., Autoregressive Integrated Moving Average (Box and Pierce, 1970), Support Vector Regression (Cortes and Vapnik, 1995)) and have been widely applied in the field of economics. For example, Tsai and Wu (2008) introduced a neural network to the bankrupt prediction. Chen et al. (2023) proposed a theory-driven deep learning method for voice chat-based customer response prediction. Moreover, Wu et al. (2021) proposed a cross-sectional machine learning approach for hedge fund return prediction and selection, and Dhar et al. (2014) applied deep learning to predict economic networks containing entities, products, and individuals. They found that the prediction methods that incorporated multiple features with high correlation were better than those that utilized a single feature. Therefore, it is important to identify factors that have a strong correlation with exchange rates to increase the accuracy of exchange rate forecasts. Aloosh and Bekaert (2022) analyzed the range of currency factors and studied the correlation between currencies. They emphasized that the values of currencies are correlated with commodity

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prices, which directly shifted our focus towards price changes in precious metals and energy. On the other hand, relying solely on single exchange rate data for forecasting can fail to utilize richer information, such as macroeconomic policies, domestic and international news, and market sentiment, which can affect short-term exchange rate fluctuations (Qiu et al., 2014). Mai et al. (2018) found that social media sentiment is a significant predictor of bitcoin's valuation and that sites like Twitter have a strong influence on future bitcoin value. Whereas, would social media have an impact on fiat currencies since it would have an impact on virtual currencies? Furthermore, given that there is a correlation between currency and commodity value, is it possible to forecast exchange rates with news concerning commodity value and other economic sectors due to news with greater rigor and professionalism than other social media (Engelberg & Parsons, 2011)? These questions motivate us to introduce news sentiment analysis into exchange rate forecasting, which is usually overlooked.

Prediction models have included both statistical models and deep learning models (Kumbure et al., 2022). Among them, Long Short-Term Memory (LSTM) and Transformer have garnered significant attention due to their inherent advantages in dealing with time series problems and have made remarkable contributions to the field of time series prediction (Wu et al., 2021; Zerveas et al., 2021; Zhou et al., 2021). Hence, we study the LSTM and Transformer by incorporating the structure of Convolutional Neural Networks (CNN), a widely used model in computer vision, that is typically used to capture local details of image features. We applied CNN to the field of exchange rate prediction, aiming to improve the sensitivity to the local features of sequences and enhance the overall prediction performance (Hoseinzade and Haratizadeh, 2019). In addition, the theory that explores the relationship between social media and equity value (Luo and Zhang, 2013) also offers valuable insights for our research on exchange rate prediction.

The theory linking social media and equity value suggests that sentiment expressed on social media platforms can have an impact on the valuation of stocks. In comparison with the stock market, the foreign exchange market exhibits a higher degree of open transactions and a greater barrier to entry. These characteristics contribute to its increased liquidity and volatility. The volatility is further amplified by its susceptibility to a broad range of macroeconomic factors and its resistance to manipulation by any single entity, making its prediction more challenging. Additionally, the high entry barrier reduces the number of participants in market discussions. Consequently, social media platforms, which are often active in stock market analysis, are less vibrant in the context of the forex market. This reduced engagement can lead to biased sentiment analysis if social media remains the primary source of information. To address this limitation, news sources are proposed as a more reliable and professional alternative to social media for forecasting forex market trends.

By incorporating insights from the theory into our research on exchange rate prediction, we can explore the possibility of leveraging social media sentiment analysis as an additional data source. This could help us gauge market sentiment, identify potential trends, and enhance the accuracy of our exchange rate forecasting models.

Our primary objective is to address two key research issues.

1. Evaluating the Impact of News Sentiment Analysis: We aim to investigate whether employing news sentiment analysis methods can effectively enhance the performance of prediction models in the context of Forex. By analyzing the sentiment expressed in news articles related to oil, gold, and Forex, we seek to determine if this information can provide valuable insights and improve the accuracy of our prediction models.
2. Assessing the Role of CNN in Improving Prediction Accuracy: Additionally, we aim to examine whether the incorporation of CNN can contribute to enhancing the prediction accuracy of both the LSTM and Transformer models. We will evaluate the effectiveness of CNN

integration in improving the performance of these models, specifically in the context of Forex prediction tasks.

By addressing these two critical issues, our research aims to uncover and highlight the potential benefits of news sentiment analysis and the role of CNN in optimizing the accuracy of prediction models for Forex forecasting. We are particularly excited about exploring the application of the Transformer model in this domain, as it has not been extensively studied before. Specifically, we constructed a financial news sentiment classification dataset and, on this basis, created a sentiment analysis model with high accuracy by fine-tuning the pre-trained BERT-WWM model.

By shedding light on the potential benefits of news sentiment analysis and the role of CNN in enhancing prediction models, our research project seeks to empower stakeholders with improved forecasting capabilities, enabling them to make informed decisions and navigate the intricate landscape of Forex trading with greater confidence.

2. Related work

The economic markets are complex systems that are influenced by a multitude of factors, such as international relations, regulatory policies, economic conditions, capital movements, and various market uncertainties. These factors can contribute to the emergence of random price fluctuations that can be difficult to predict.

To tackle this challenging problem, researchers have employed various approaches to forecasting. To evaluate the reliability and validity of our research, we compared and discussed some related works. The majority of economic forecasting works have focused on stocks and exchange rates, and they concluded that it is quite effective to integrate techniques of sentiment analysis into forecasting methodologies for predicting stock prices (Kumbure et al., 2022).

To summarize, our method for analyzing news sentiment employed the BERT-WWM model, a leading-edge development in Transformer architecture, tailored for Chinese text processing. This choice was instrumental in crafting a sentiment analysis model adept at categorizing financial news. We also delved into the benefits of the Transformer framework, particularly its proficiency in handling long-range dependencies, and we were among the first to apply it in exchange rate prediction.

Further refining this model, we integrated CNN modules, drawing inspiration from the theory of receptive fields. This integration aimed to harness the Transformer's strength in capturing extensive interdependencies and augment it with the feature extraction capabilities of CNNs, thus forging a more potent model for predicting exchange rates.

Our work builds upon previous studies that blended CNNs with LSTMs, but with a significant advancement. We adapted the Transformer architecture to meet the specific demands of financial news sentiment analysis, thereby exploring and enhancing its utility in forecasting exchange rates. This approach not only leveraged the advancements in Transformer technology but also tailored it to excel in the nuanced field of financial news analysis.

2.1. Sentiment analysis

For traditional sentiment analysis methods, manual feature extraction is performed first, and then Support Vector Machine (SVM), Naive Bayes (NB), etc. are used for classification. Pang et al. (2002) introduced multiple sentiment features and used machine learning methods (NB and SVM) to generate a more accurate solution for sentiment classification. However, compared with these traditional methods, deep learning has made breakthrough in recent years.

Word embedding is a popular technique for representing the vocabulary of a document, mapping textual words into vector representations. Word2Vec is a well-known technique for word embeddings,

utilizing a shallow neural network to process a text before passing it into a deep learning network (Mikolov et al., 2013). Furthermore, it is effective for improving word embedding to draw on prior knowledge. Li et al. (2016) proposed a knowledge-enabled textual analysis approach for financial market surveillance. Yang et al. (2022) used the similar approach to analyze firm reports. After the word embedding step, all word vectors need to be integrated for sentiment analysis through a specific model. Huang et al. (2017) and Hassan and Mahmood (2017) have proposed architectures combining CNN and LSTM for classifying short texts on top of Word2Vec. Zhang, Wei, Zheng, Li, and Zeng (2022) took advantage of local and long-term word relations on social media by integrating CNN and LSTM. Furthermore, Nguyen et al. (2015) utilized sentiment analysis on social media for stock movement prediction.

Vaswani et al. (2017) proposed the Transformer model, which utilizes the attention mechanism to obtain global structural information and perform parallel computing. The ELMo model (Peters et al., 2018), which is based on bidirectional LSTM, brings a text representation method based on context information.

Based on the Transformer model, researchers have developed pre-trained language models such as BERT (Devlin et al., 2019), ERNIE (Zhang et al., 2019), which have shown superior performance in many NLP tasks compared to traditional models. Among them, BERT has achieved excellent results in numerous downstream NLP tasks, promoting the development of NLP technology into a new era. In 2019, the Joint Laboratory of HIT and iFLYTEK Research released a BERT-WWM Chinese pre-training model that has demonstrated superior performance compared to other Chinese pre-training models Cui et al., (2021).

2.2. Prediction model

In the domain of exchange rate forecasting, the primary method for predicting future trends involves analyzing sequences of historical exchange rate data. Prior research in this field has concentrated on various modifications and adaptations of predictive models, aiming to improve their effectiveness specifically for exchange rate forecasting.

Selvin et al. (2017) compared CNN with ARIMA, LSTM, and RNN for stock price prediction. The results showed that deep learning models greatly outperformed ARIMA, and CNN has a better ability to capture change patterns. Hoseinzade and Haratizadeh (2019) introduced 3D-CNN into stock price prediction, and their experiments suggested that CNN managed to exploit information from various sources and especially outperformed traditional technical methods, which used artificial indicators to reveal more valuable information within raw data. Alonso-Monsalve et al. (2020) exploited CNN-LSTM for high-frequency trend prediction of cryptocurrency exchange rates. Additionally, Livieris et al. (2020) utilized 2 versions of CNN-LSTM in the task of gold price prediction. Yan et al. (2021) used CNN-LSTM to predict air quality index. They all successfully transferred CNN-LSTM to their own specific fields, which strongly inspired us to utilize CNN-LSTM to forecast exchange rates. However, it is still necessary to scientifically and reasonably adjust its structure to cater to the unique characteristics of exchange rate data.

Unfortunately, LSTM have limitations in effectively utilizing information from earlier time. In the context of forex forecasting, accounting for a longer historical time span is crucial for reducing prediction errors. This limitation underscores the need to introduce the Transformer model in this field. There have been several great models based on Transformer architecture for long-term time series prediction, such as Informer (Zhou et al., 2021), Autoformer (Wu et al., 2021) and they achieved excellent results in long-term time-series forecasting.

Additionally, utilizing multimodal methods for prediction is an emerging trend. Corizzo and Rosen (2023) proposed a novel stacked ensemble approach for stock market prediction, which takes into account news headlines, multivariate time series data, and multiple base models as predictive factors. Zou and Herremans (2023) introduced a multimodal deep learning model designed to predict extreme Bitcoin

price fluctuations. This model uses candlestick data, prices of various related assets, technical indicators, and Twitter content as inputs.

3. Methodology

Fig. 1 gives an overview of our work. The sentiment analysis model is responsible for transforming the daily news text into an overall sentiment direction for the daily market. This result is then combined with exchange rate data and forwarded to the forex forecasting model. The forex forecasting model, leveraging rich feature data, autonomously establishes connections between features, delves into deep information, disregards irrelevant details, and ultimately makes accurate predictions for future exchange rate data.

3.1. Establishment for sentiment analysis model

3.1.1. Sentiment analysis model: BERT-WWM

The BERT model based on the Transformer architecture shows significant advantages for better extracting the sentiment features of news by comparing several sentiment analysis models (Devlin et al., 2019). Fig. 2 presents our BERT model. The right side is the Transformer Encoder, whose multi-head attention mechanism completely exploits features between tokens. It is also a key component of the left BERT model and our other prediction model in the following section. However, the Transformer Encoder is not sensitive to the order of tokens because of a multi-head attention mechanism. In order to complement the order information between tokens, the positional encoding should be introduced.

The potent capabilities of BERT in natural language processing are credited to its innovative pretraining mechanism, namely the Masked Language Model (MLM) task. In the initial phase, learning corpora are obtained by randomly replacing words in input text with a special [MASK] token, as illustrated in the first two rows of Table 1. Subsequently, the BERT model endeavors to predict these masked words utilizing contextual information provided by neighboring words. This bidirectional approach, akin to cloze-style tasks in human language learning, enables BERT to simultaneously consider both left and right context, leading to a nuanced understanding of word semantics and context.

However, the strategy of masking words in the BERT pre-training stage is not very appropriate for Chinese. Due to language differences, the masking unit of BERT for Chinese can only be a character, but the smallest unit of information in Chinese is a word that contains more characters. Moreover, there is no obvious division between each Chinese word in a sentence, while in English, such separation is indicated by the space character. In certain situations, multiple words in English may correspond to a single word in Chinese. For instance, in Table 1, the combination of “Gold” and “Market” precisely forms a common word in Chinese. Therefore, for Chinese, a more effective masking approach would be to simultaneously mask the words “Gold” and “Market”. Based on the above analysis, the BERT-WWM (Cui et al., 2021) is a better choice for multiple downstream tasks of Chinese natural language processing due to its pre-training strategy of masking the whole word instead of a character. Table 1 shows a comparison of the two masking mechanisms. Note that the original corpus is in Chinese, and Table 1 has been translated into English for explanatory purposes.

Consequently, for our task of sentiment analysis on Chinese news, we adopted the pre-trained BERT-WWM (huggingface.co/hfl/chinese-bert-wwm-ext) instead of BERT as the backbone of our model. Pre-trained models typically come with fixed hyperparameters, so there is no need for hyperparameter optimization. We can simply fine-tune the model for downstream tasks. In our task, we replace the header layer with a dense layer that outputs three elements. These indicate the probabilities of three sentiment categories (Positive, Neutral, and Negative). The category to which the maximum probability belongs is the final classification result. The formula describing the model inference is as follows:

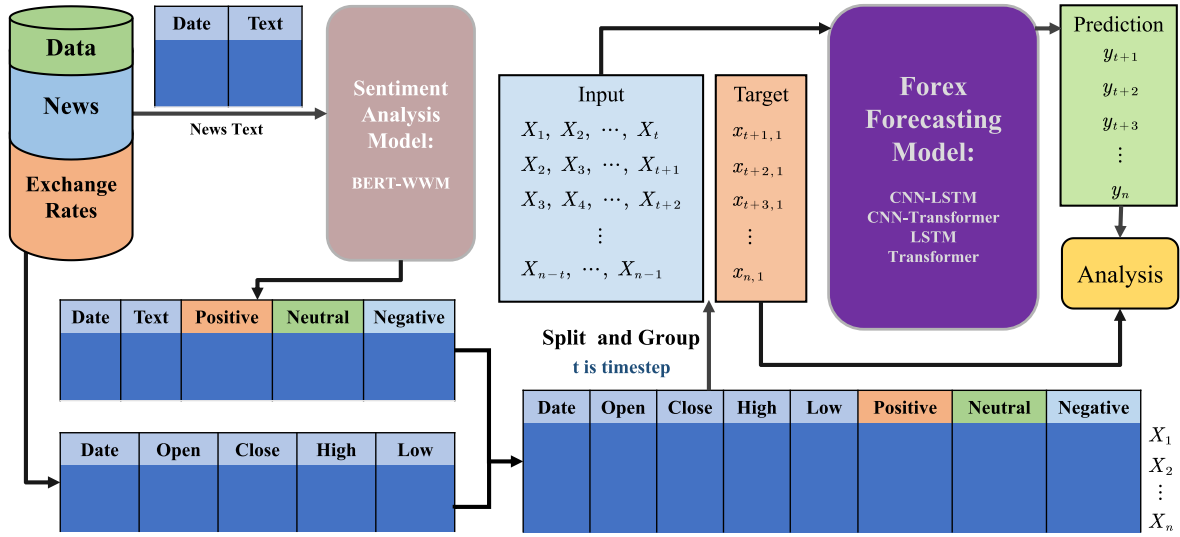


Fig. 1. Overview of workflow and data operations.

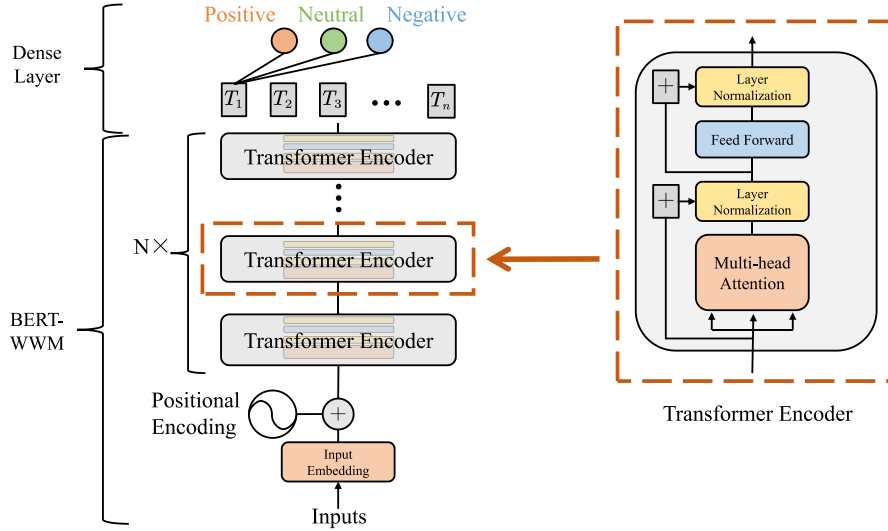


Fig. 2. Our bert model fine-tuned from BERT-WWM

Table 1

Two different masking mechanisms: An example.

Sentence	Gold market analysis: It is expected that the gold market outlook will be a rhythm of small drops and big rises
BERT	Gold [MASK] analysis: It is expected that the gold [MASK] outlook will be a rhythm of small [MASK] and big rises
BERT-WWM	[MASK] [MASK] analysis: It is expected that the [MASK] [MASK] outlook will be a rhythm of [MASK] [MASK] [MASK] [MASK] [MASK]

First, process the raw input data $X_{n \times m}$, where pos is the position in the length m of sentence, i is the index in the dimension n of embedding space and d_{model} is a hyperparameter. P is the data input to the Transformer Encoder.

$$\text{Positional - Encoding}(pos, i) = \begin{cases} \sin(pos/10000^{i/d_{model}}), & i \text{ is even} \\ \cos(pos/10000^{i/d_{model}}), & i \text{ is odd} \end{cases} \quad (1)$$

$$p_{i,j} = \text{Positional - Encoding}(j, i) + x_{i,j} \quad (2)$$

After P is calculated by multiple layers of Transformer Encoder, it

will get the result T . Then, extract the first column element of T , perform a projection transformation, and obtain the sentiment vector S . Finally, based on the sentiment vector, derive the classification result y .

$$T_{n \times d_{model}} = \text{Transformer}^n(P) \quad (3)$$

$$S = T_1 W \quad (4)$$

$$y = \underset{i}{\operatorname{argmax}}(s_i) \quad (5)$$

We also used the sentiment vector to refine the results of the prediction models.

3.1.2. Dataset for sentiment analysis

The news we collect is from fx678.com, which is the largest professional foreign exchange portal in China and covers news reports, expert commentary, and related market information on forex, gold, and oil. Considering the daily volume distribution and completeness of the news, we have selected news from December 2018 to December 2022 in forex, gold, and oil. The news is labeled into three categories based on whether the news has a positive, negative, or neutral impact on the market.

Specifically, the labeling approach relies on manual judgment and is checked by the lexicon-based approach (Shapiro et al., 2022). During the initial labelling, each news item is labeled by two individuals with expertise in finance. If their judgments are inconsistent, the news item is subjected to a second round of manual judgment by a committee. The committee consists of five professionals in finance, who discuss and decide the final result.

For the training and evaluation of the sentiment classification model afterwards, we divided the news from December 2018 to March 2021 into the training set and the news from April 2021 to December 2022 into the test set. As the model we employed is pre-trained, the hyperparameters of pre-trained models are typically fixed, and there is no need for hyperparameter optimization. Consequently, we did not partition a validation set on this dataset. The information in the dataset is shown in Figs. 3–5.

Fig. 3 illustrates the daily fluctuations in the quantity of three types of news in the dataset throughout 2021. The fluctuations remain generally consistent, with the number of exchange rate news typically exceeding that of oil and gold news. Figs. 4 and 5 present the characteristics of the training set and test set. Fig. 4 primarily showcases the word frequency distribution, revealing frequent occurrences of terms related to the economies of China and the United States in both the test and training sets. This prevalence contributes to the potential for predicting exchange rates between China and the United States. Note that the original corpus is in Chinese, and labels in Fig. 4 has been translated into English for explanatory purposes. Additionally, information about other countries, such as the United Kingdom, is also frequently present. Fig. 5 displays the distribution of news sentiment in the training and test sets, indicating a generally similar distribution between the two sets, albeit with a slight deviation in the distribution of positive and negative news.

3.1.3. Fine-tuning and evaluation

Fine-tuning is a training strategy conducted after the pretraining phase of the BERT-WWM model. At the end of pretraining, BERT-WWM has learned general language representations from a large-scale Chinese corpus, but it has not been optimized for specific tasks. The fine-tuning process essentially involves adjusting the output layer of the model to meet task requirements. Subsequently, the model is further trained on a specific small-scale dataset by tuning model parameters (often with few training epochs). This transfer learning approach enables the model to exhibit excellent performance in downstream specific tasks.

In the actual fine-tuning process, we added a dense layer to the backbone of BERT-WWM, aiming to transform the output of BERT-WWM into a three-dimensional output corresponding to sentiment classification. Simultaneously, we allow the parameters of both the backbone BERT-WWM and the dense layer to be updated concurrently, indicating that our fine-tuning adapts BERT-WWM more specifically to the sentiment classification task. The fine-tuning process consists of 100 training epochs with a batch size set to 64. The model ultimately

achieves an accuracy of 91.2% on the test set, and Fig. 6 illustrates the confusion matrix of the model on the test set.

Through the confusion matrix, we can observe that the model performs well on the “Positive” and “Negative” classes, but encounters some errors in the “Neutral” class. However, it is noteworthy that despite the relatively higher misclassification rate for the “Neutral” class, the misclassifications are roughly evenly distributed between “Positive” and “Negative” classes (“Positive” and “Negative” are respectively 680 and 692). From a conditional probability perspective, when the model encounters a sample labeled as “Neutral” and a misclassification occurs, the probabilities of misclassifying as “Positive” and “Negative” are both 0.5. This aligns with the inherent meaning of the term “Neutral”, as if we were to quantify “Neutral”, it would naturally be considered as an equal mix of “Positive” and “Negative”.

Upon further consideration, to more thoroughly examine whether the model understands the true meaning of “Neutral”, we isolate samples labeled as “Neutral” but misclassified by the model. We inspect the model’s sentiment vectors for these all samples, as shown in Table 2. Despite the model’s misclassification, the difference in sentiment vectors between “Positive” and “Negative” is within 5%, indicating that the model itself tends to vacillate between “Positive” and “Negative” choices, confirming the model’s accurate understanding of “Neutral”.

In summary, we believe that our sentiment analysis model has grasped the essence of sentiment analysis tasks, and sentiment vectors better reflect the true sentiment orientation of news than classification results.

3.2. Establishment for forex forecasting model

3.2.1. Forex forecasting model

The model processes a sequence containing data for several days, defining each day’s data as a unit. Assuming the sequence comprises L units, where L is the length of the sequence, also referred to as the timestep. Furthermore, each unit contains multiple features, allowing us to view the unit as a column vector, with each dimension of the column vector corresponding to a feature. As the sequence contains multiple units, it can be regarded as a stack of multiple column vectors. Thus, the sequence is essentially an $M \times L$ matrix, where M is the number of features, and L is the sequence length (or timestep). The output of models is a scalar, representing a open price in the unit for the day following the last day in the input sequence. Figure 7 shows the above process.

3.2.1.1. CNN-LSTM. The LSTM, an effective model for dealing with sequence prediction problems, is made up of a memory cell and three gates: a forget gate, an input gate, and an output gate. Fig. 8 presents the LSTM architecture.

The forget gate determines how much information from both input data and previous states should be forgotten in the memory cell by the sigmoid function $\sigma(\cdot)$. The formula is shown below:

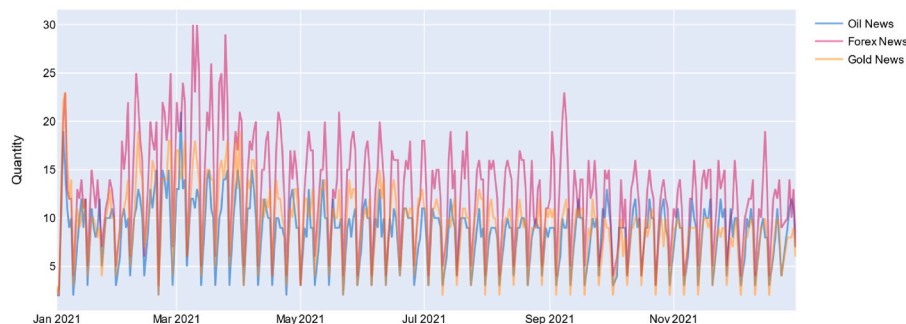


Fig. 3. Number of daily news about three types in 2021.

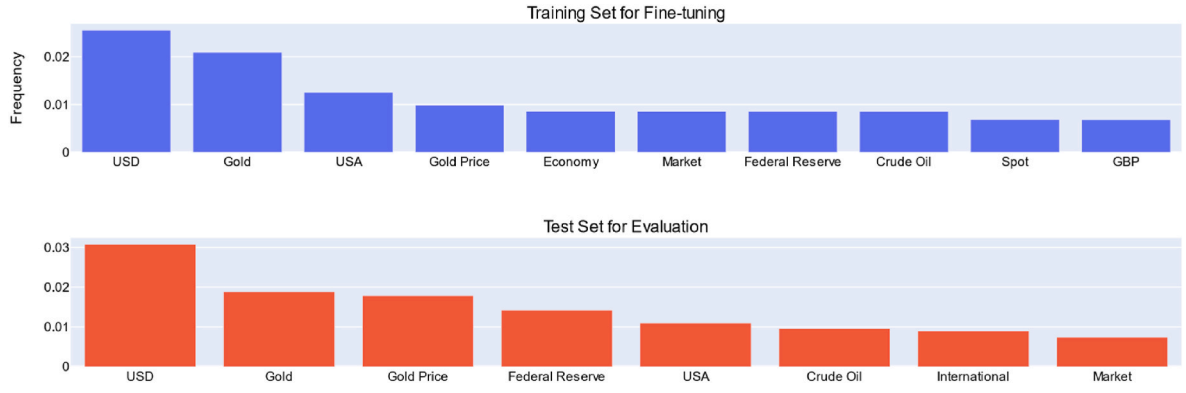


Fig. 4. Word frequency across all news articles.

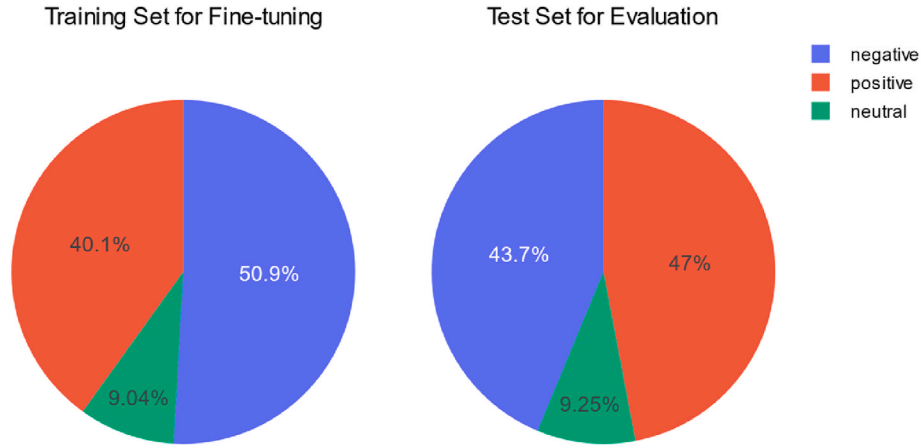


Fig. 5. News sentiment distribution.



Fig. 6. Confusion Matrix on Test set.

Table 2

Sentiment vectors of samples labeled as neutral but misclassified: Examples.

Positive	Neutral	Negative
0.244723096	-0.044441628	0.259985113
0.273960769	-0.163757557	0.259719908
0.192315608	-0.372470403	0.170739238

$$F_t = \sigma(W_f[H_{t-1}, X_t] + b_f) \quad (6)$$

where W_f represents weight and b_f represents bias.

In the same way, the input gate determines how new information should be integrated into the memory cell. The formula is as follows:

$$I_t = \sigma(W_i[H_{t-1}, X_t] + b_i) \quad (7)$$

To update the memory cell C_t , it requires $\tilde{C}_t$ and I_t . These formulas

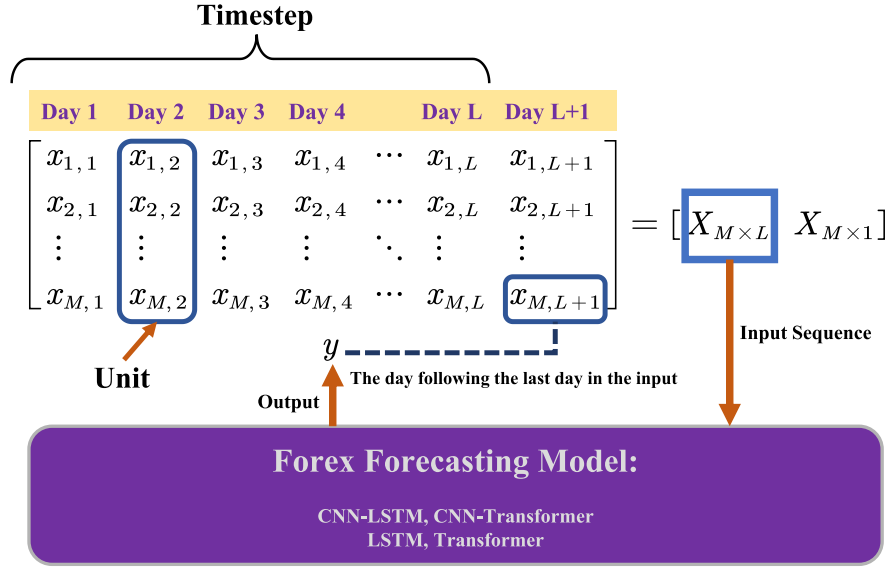
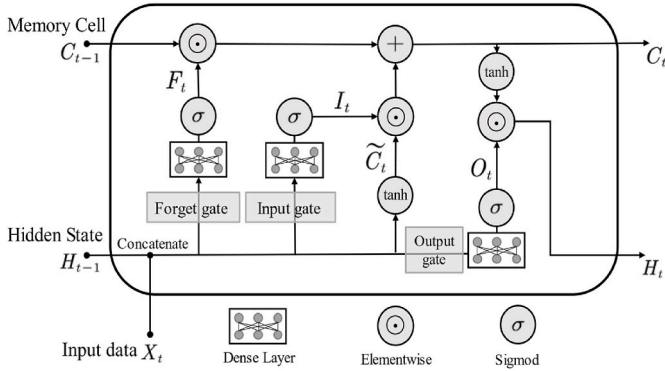


Fig. 7. Model input and output.



are shown below:

$$\tilde{C}_t = \tanh(W_c[H_{t-1}, X_t] + b_c) \quad (8)$$

$$C_t = F_t \odot C_{t-1} + I_t \odot \tilde{C}_t \quad (9)$$

The hidden state is the main source of information for the final output A_t . To update the hidden state H_t , it requires O_t and C_t . These formulas are shown below.

$$O_t = \sigma(W_o[H_{t-1}, X_t] + b_o) \quad (10)$$

$$H_t = O_t \odot \tanh(C_t) \quad (11)$$

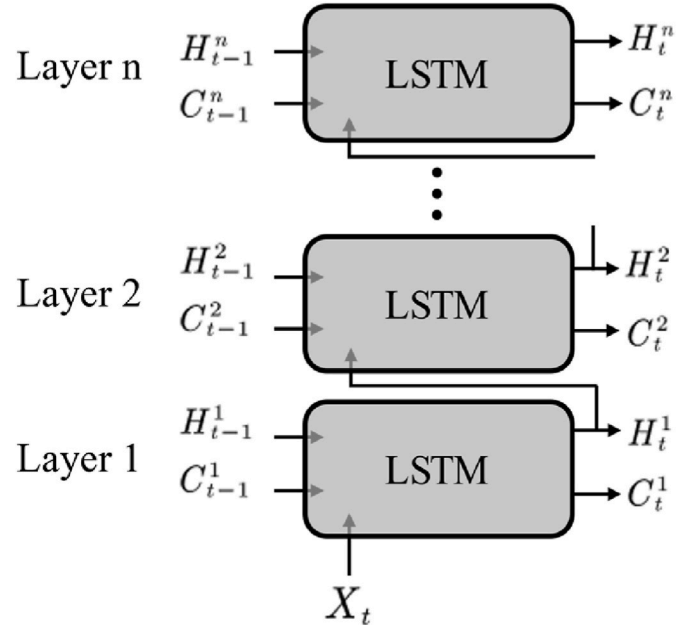
In the formula below, resizing the weight matrix W_a allows the dimension of the final output A_t to match the requirements.

$$A_t = W_a H_t + b_a \quad (12)$$

In general, the stacked LSTM is an extremely effective method for complex time series tasks, which takes the hidden state of the previous layer as the input data for the current layer. Fig. 9 shows the stacked LSTM architecture.

While the memory cell has enhanced LSTM performance in tasks with long time steps, there is still a need to improve LSTM's ability to capture features at distant time points and global information.

Due to the recurrent nature of LSTM, a type of recurrent neural



network architecture that inherently suffers from the issue of gradient instability, the model faces challenges in effectively leveraging temporally distant data. Assuming LSTM can learn dependencies up to a maximum timestep denoted as T , with an input sequence timestep of L , and the model tasked with predicting the unit at $L+1$, LSTM can only effectively learn data from $[L-T+1, L]$ in the input sequence. Data indexed in $[1, L-T]$ poses difficulties for LSTM due to gradient instability, hindering the learning of features from this segment, despite it may significantly contribute to the prediction.

In light of this analysis, our goal is to empower LSTM to learn from data at $[1, L-T]$. Drawing inspiration from the idea in CNN, where receptive fields expand with additional layers, we propose using CNN to aggregate units in the sequence, thereby shortening the sequence length. Note that the convolutional kernel has one dimension fixed to match the feature size of input sequence, with flexibility in another dimension, which is commonly referred to as a 1D convolutional kernel. The rele-

vant formulas are as follows.

- L : Length of original sequence.
- M : Feature size in the original sequence.
- $\tilde{L}$: Length of sequence after CNN aggregating.
- P : Padding amount in the direction of L .
- K : Kernel size is $M \times K$.
- N : Number of kernels.
- g_i : Convolution operation of the i^{th} kernel
- S : Stride of the convolutional kernel.

$$\text{CNN}(X_{M \times L}) = \begin{bmatrix} g_1(X_{M \times L}) \\ \vdots \\ g_N(X_{M \times L}) \end{bmatrix} = \begin{bmatrix} \dot{X}_{1 \times \tilde{L}}^1 \\ \vdots \\ \dot{X}_{1 \times \tilde{L}}^N \end{bmatrix} = \dot{X}_{N \times \tilde{L}} \quad (13)$$

$$\tilde{L} = (L + 2P - K) / S \quad (14)$$

By subjecting the sequence to multiple layers of CNN, we aim to reduce the sequence length to be within LSTM's learnable timestep T . The sequence, after CNN processing, is then fed into LSTM. Analogous to the analysis of receptive fields in CNN, in this scenario, receptive field of LSTM can expand up to the entire input sequence.

Consequently, we introduce a hybrid model, CNN-LSTM, which utilizes CNN to extract features from raw data into high-dimensional space and subsequently feeds the higher-dimensional feature data into LSTM for prediction. Fig. 10 illustrates the architecture of CNN-LSTM.

To ensure that the convolutional output $g_i(\cdot)$ is a vector, the width M of the convolution kernel is matched with the number of rows in the input. This setup also allows for variations in the length K of the convolution kernel, enabling it to aggregate information over different time intervals, thus enhancing the feature set. The convolutional output vectors are subsequently stacked to form the new sequence (or matrix) $\dot{X}_{N \times \tilde{L}}$, where each row corresponds to the output $\dot{X}_{1 \times \tilde{L}}^i$ of an individual convolution kernel. To further expand the feature set, increasing the number of convolution kernels is an option, though this also increases the size and computational load of the convolutional output. Outputs from multiple layers of the CNN are fed into the LSTM recurrent network. After processing in the LSTM, the hidden state from its top layer is inputted into the dense layer to produce the final result. It is noteworthy that to alleviate overfitting and improve the model's generalizability, the dropout mechanism is frequently utilized in both the LSTM and dense layers.

In summary, the CNN module is employed for the extraction of raw data and the enrichment of features, similar to feature engineering in machine learning. The LSTM module is used for data prediction, while the dense layer performs feature space projection and transformation to extract the final answer.

3.2.1.2. CNN-transformer. The Transformer architecture is capable of handling tasks associated with time series, thanks to its multi-head attention mechanism that establishes deep connections between data points at different timestamps. Fig. 11 demonstrates how the Transformer architecture is applied to time series analysis, with the Transformer Encoder being the main core of the entire model on the left. The detailed structure of the Transformer Encoder is depicted on the right. It is noteworthy that the original Transformer Encoder, applied in the natural language processing domain, handles samples of varying lengths, typically employing padding mechanisms to equalize all sample lengths. However, in such cases, traditional Batch Normalization becomes less effective due to instability in variance and mean, influenced by padding values (usually 0), and fails to adequately address covariate shift. Therefore, the original Transformer Encoder utilizes Layer Normalization instead of Batch Normalization.

In the context of exchange rate prediction, where each training data point is of consistent length and does not require padding, the variance among features may significantly contribute to the final outcome. Layer normalization could diminish this variance, potentially reducing prediction accuracy. Hence, we continue to use traditional Batch Normalization to better suit exchange rate prediction. Additionally, we posit that the information density in exchange rate data is lower than in natural language, and given that the Transformer architecture originated from natural language processing, its powerful fitting capacity might lead to training instability on the relatively simple exchange rate data. This circumstance further necessitates the use of Batch Normalization to stabilize training (Zerveas et al., 2021).

In Fig. 11, for the left prediction model, the input data $X_{n \times m}$ is first reshaped by projection, then summed with the positional encoding to get P , and finally input P into the Transformer Encoder. The outputs of the last Transformer Encoder are concatenated into a matrix Z and projected to the final result Y through a dense layer. The related formulas are shown below.

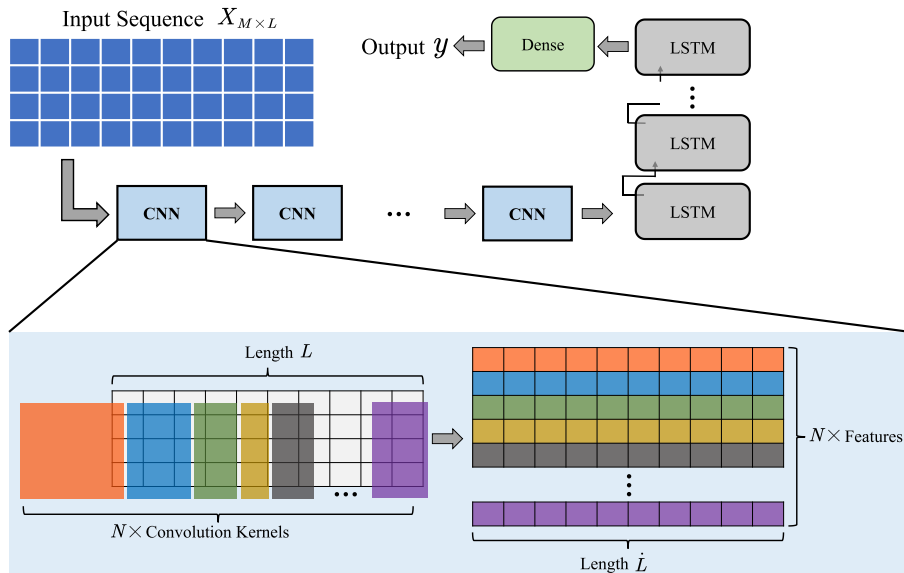


Fig. 10. CNN-LSTM architecture.

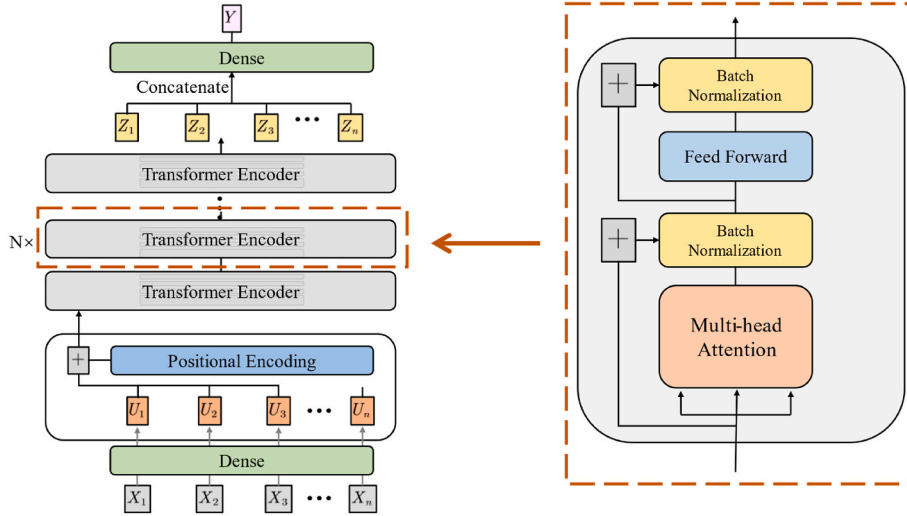


Fig. 11. Transformer architecture for time series.

$$\text{Positional - Encoding}(pos, i) = \begin{cases} \sin(pos/10000^{i/d_{model}}), & i \text{ is even} \\ \cos(pos/10000^{i/d_{model}}), & i \text{ is odd} \end{cases} \quad (15)$$

$$U_{n \times m} = W_{n \times n} X_{n \times m} \quad (16)$$

$$p_{i,j} = \text{Positional - Encoding}(j, i) + u_{i,j} \quad (17)$$

First, process the raw input data $X_{n \times m}$, where pos is the position in the timestep m , i is the index in the features n , d_{model} is a hyperparameter, P is the data input to the Transformer Encoder.

After P is calculated by multiple layers of Transformer Encoder, it will get the result Z .

$$Z = \text{Transformer}^n(P) \quad (18)$$

In the Transformer Encoder, the multi-head attention module exhibits a time complexity of $\Theta(L^2)$. Consequently, the model with stacked Transformer Encoders faces significantly longer training times due to this high complexity. To reduce the total runtime of the model while ensuring stable accuracy, it is imperative to decrease the number of Transformer Encoder layers and more importantly, reduce the sequence length to cut down the computational time of the multi-head attention module.

Therefore, we introduce the CNN-Transformer model, which incorporates a CNN module akin to that used in CNN-LSTM architectures. This model employs CNN modules to replace some of the bottom Transformer Encoder layers, aiming to reduce computation time. Additionally, the CNN module aggregates local information, thereby shortening the sequence length. It significantly decreasing the computational time for subsequent Transformer Encoder layers.

Assuming there are n CNN modules, each with identical hyperparameters (K , P and S). Let's denote the input sequence length of the i^{th} CNN module as L^i and the output sequence length as $\tilde{L}^i$. It's evident that $\tilde{L}^i = L^{i+1}$. This assumption leads to the following derivation.

$$L^{i+1} = (L^i + 2P - K) / S + 1 \quad (19)$$

$$L^{i+1} + \frac{K - 2P - S}{S - 1} = \frac{1}{S} \left(L^i + \frac{K - 2P - S}{S - 1} \right) \quad (20)$$

$$L^{n+1} = L^1 \cdot S^{-n} + (S^{-n} - 1) \cdot \frac{K - 2P - S}{S - 1} \quad (21)$$

Obviously, the length of the input sequence decreases exponentially with the increasing number of CNN modules, significantly reducing the

sequence length that the subsequent Transformer Encoder needs to process. It is also crucial to note that the Transformer Encoder has a substantial number of parameters, which can easily lead to overfitting. To mitigate this issue, Dropout mechanisms are extensively utilized in each Transformer Encoder to enhance the model's generalizability. Fig. 12 shows the architecture of the CNN-Transformer model.

3.2.2. Forex dataset

The forex data is from [Investing.com](https://uk.investing.com/currencies/) (uk.investing.com/currencies/), a financial website in the world. We collected data on the USD/CNY, EUR/CNY and GBP/CNY currencies from December 2018 to December 2022, including close, open, high and low prices. We divided the forex data from December 2018 to August 2020 into the training set, the data from September 2020 to March 2021 into validation set, and the data from April 2021 to December 2022 into the test set. The mean and variance of everyday exchange rate prices are shown in Table 3.

To enrich market features in the exchange rate dataset, it's necessary to integrate market sentiment information extracted from previous news datasets. First, we apply a softmax function to the sentiment vectors S_{news} of each news item (derived from the output of BERT-WWM), resulting in normalized outcomes P_{news} . Then, we aggregate the normalized results of all news items for each day to form a sentiment feature vector T_{day} that reflects daily market sentiment. The specific formulas are as follows. Finally, we align and concatenate exchange rate data with market sentiment data based on date information. This process yields three datasets featured with comprehensive market information and exchange rate fluctuations. Table 4 illustrates the dataset created by concatenating the USD/CNY exchange rate with market sentiment information. Additionally, datasets for the EUR/CNY and the GBP/CNY exchange rates have also been concatenated respectively with integrated market sentiment information.

$$S = [s_1, s_2, s_3]^T \quad (22)$$

$$P_{news} = \text{softmax}(S_{news}) = [p_1, p_2, p_3]^T \quad (23)$$

$$p_i = \exp(s_i) / \sum_{k=1}^3 \exp(s_k) \quad (24)$$

$$T_{day} = \sum_{k=1}^n P_k \quad (\text{assuming there're } n \text{ news item on the day}) \quad (25)$$

Fig. 13 details the above process, where the exchange rate datasets without market sentiment information is utilized for subsequent

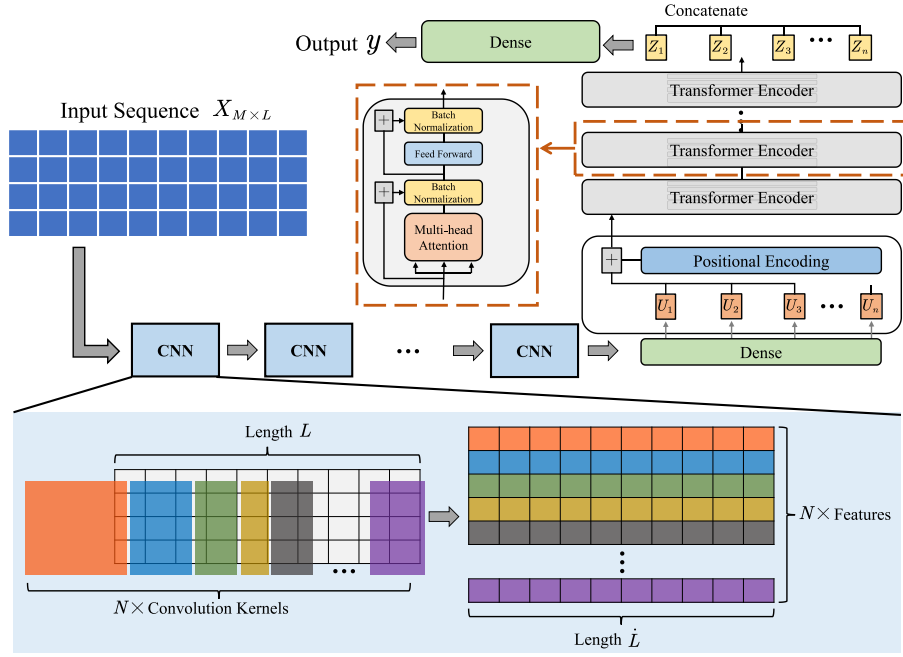


Fig. 12. CNN-transformer architecture.

Table 3

Statistics of the forex dataset.

	Close			Open			High			Low		
	USD	EUR	GBP	USD	EUR	GBP	USD	EUR	GBP	USD	EUR	GBP
Mean	6.750	7.582	8.711	6.751	7.585	8.714	6.762	7.612	8.753	6.740	7.559	8.674
Standard Deviation	0.267	0.345	0.303	0.267	0.345	0.302	0.269	0.342	0.296	0.265	0.347	0.308

Table 4

USD/CNY dataset with market sentiment: Part items.

USD/CNY				News Sentiment			Date
Close	Open	High	Low	Positive	Neutral	Negative	
6.3839	6.391	6.4019	6.3808	6.778	10.057	11.813	10.22.2021
6.3925	6.394	6.4031	6.3872	4.806	10.145	11.709	10.21.2021
6.3936	6.3818	6.3979	6.3818	6.399	10.134	12.148	10.20.2021
6.3822	6.4299	6.4299	6.3797	4.455	10.182	11.618	10.19.2021

comparative experiments.

4. Experiments

The output feature of models is “open price”. The best performance on the validation set of each forex prediction model is searched within a fixed range of hyperparameters. The hyperparameters are shown in Table 5 and Table 6. The formulas of the evaluation metrics are shown below.

$$MSE = \frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - y_i)^2 \quad (26)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\tilde{y}_i - y_i| \quad (27)$$

$$ROC = \frac{MAE_{old} - MAE_{new}}{MAE_{old}} \quad (28)$$

The performance of the models is measured by mean square error

(MSE) and mean absolute error (MAE), where $\tilde{y}$ is the model output and y is the ground truth. To quantify the improvement in model prediction error resulting from the addition of market sentiment information to the dataset or the integration of CNN modules, we introduce a rate of change metric (ROC). Here, MAE_{old} represents the value before the improvement and MAE_{new} denotes the value after the improvement. It is noteworthy that MSE serves as the loss function for training all the forex forecasting models and all input data are subjected to zero-mean standardization during the experiments.

To examine whether the integration of CNN can enhance the predictive accuracy of LSTM and Transformer models, we initially conduct a search within the scope of Tables 4 and 5. Based on the performance of the four models on the validation set, we determine the best hyperparameters respectively for each model. Subsequently, we compare the test results of these models configured with the best hyperparameter. This comparison aims to observe whether the integration of CNN modules affects the final predictive accuracy of the models. Note that the forex dataset used in the above experiment includes market sentiment.

To verify whether introducing market sentiment information indeed optimizes prediction results, we duplicated the forex dataset and

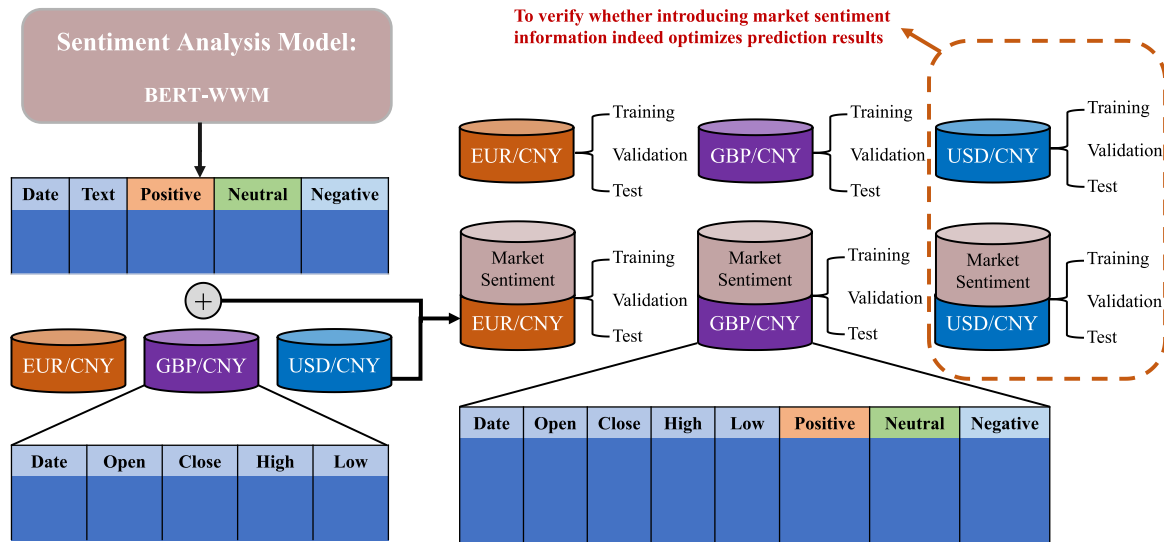


Fig. 13. Forex dataset processing workflow.

Table 5
Hyperparameters of CNN-LSTM and LSTM.

CNN-LSTM		LSTM
Number of LSTM Layers	1, 2, 3	1, 2, 3
Hidden State Size	128, 256	128, 256
Number of Layers in CNN	1, 2	\
Kernel Size	3, 5	\
Padding	0, 1, 3	\
Stride	1, 3, 5	\
Time Step	9, 12, 21, 33, 54, 87, 96, 105, 117, 126, 135, 141	9, 12, 21, 33, 54, 87, 96, 105, 117, 126, 135, 141

Table 6
Hyperparameters of CNN-transformer and transformer.

CNN-Transformer		Transformer
Number of Transformer Encoder Layers	1, 2, 3	1, 2, 3
Hidden State Size	128, 256	128, 256
Number of Heads	2, 4, 8	2, 4, 8
Number of Layers in CNN	1, 2	\
Kernel Size	3, 5	\
Padding	0, 1, 3	\
Stride	1, 3, 5	\
Time Step	9, 12, 21, 33, 54, 87, 96, 105, 117, 126, 135, 141	9, 12, 21, 33, 54, 87, 96, 105, 117, 126, 135, 141

removed its market sentiment information. This process created a basic comparison dataset that only contains exchange rate fluctuation data. The partition ratio of the comparison dataset remains unchanged. Identical experimental procedures are conducted on both datasets, including selecting optimal hyperparameters based on the models' performance in the validation set and assessing the models' actual performance on the test set. By comparing the test results of the same models on different datasets, we can validate the effectiveness of integrating market sentiment into the analysis.

5. Results and discussion

5.1. Effectiveness of integrating CNN modules

After extensive searching and training, Table 7 displays the best hyperparameters for each model under different validation sets. It also

shows the test results of models equipped with these best hyperparameters.

Based on Tables 7 and it's clear that incorporating CNN modules into LSTM and Transformer models is highly effective. This is further evidenced by the considerable ROC values. While there are instances in the EUR dataset where CNN-LSTM underperforms compared to LSTM, the ROC value indicates that this degradation is not severe. These exciting results preliminarily confirm the validity of our idea to integrate CNN modules into LSTM and Transformer models. To further substantiate our hypothesis that CNN, by shortening sequence lengths, incorporates effective predictive information into the learnable timestep of LSTM, we attempt to change the timestep in the models equipped with the best hyperparameters, observing the models' sensitivity to timestep alterations.

Fig. 14 clearly depicts the performance of models equipped with the best hyperparameters across three exchange rate datasets for different time steps. The section highlighted in purple represents its timestep is greater than 55. In this section, the performance gap between CNN-LSTM and LSTM gradually widens, followed by fluctuations, and eventually, the gap narrows again. The trend of both red and blue curves remains highly similar throughout.

As the timestep increases, the LSTM model struggles with longer input sequences, leading to unstable gradients and an inability to utilize distant data. This overlooked data, potentially valuable, is accumulated, resulting in longer input sequences that do not provide LSTM with more useful information. Instead, they increase training time and reduce training stability. However, the CNN-LSTM model behaves very differently. The CNN module exponentially shortens the length of sequences, easily bringing even lengthy inputs within the learnable timestep of LSTM. Longer input sequences offer more valuable data for CNN-LSTM, and the CNN module ensures that all valuable information can be learned for LSTM. Thus, the difference between LSTM and CNN-LSTM becomes more pronounced with increasing timesteps. Overall, our findings suggest that introducing CNN modules effectively compresses more information into the limited learnable timestep of LSTM, allowing it to learn a broader range of features.

5.2. Effectiveness of news sentiment analysis

The comparative experiment results of the models on three different exchange rate datasets are shown in Table 8. In the table, "w" stands for "with," indicating the datasets include market sentiment information, while "o" represents "without," meaning the datasets exclude market

Table 7

Best hyperparameters and performance.

Dataset	Model	CNN-Transformer	Transformer	CNN-LSTM	LSTM
USD	Hyperparameters	Hidden size = 256	Hidden size = 256	Hidden size = 256	Hidden size = 256
		Timestep = 87	Timestep = 33	Timestep = 141	Timestep = 135
		Number of Transformer Encoder Layers = 1	Number of Transformer Encoder Layers = 1	Number of LSTM Layers = 3	Number of LSTM Layers = 3
		Number of Heads = 4	Number of Heads = 4	Number of CNN Layers = 1	
		Number of CNN Layers = 1		Kernel Size = 3	
		Kernel Size = 3		Padding = 0	
EUR	Hyperparameters	Padding = 0		Stride = 3	
		Stride = 3			
		Result			
		MSE	0.07101	0.00966	0.00963
		MAE	0.20757	0.06843	0.07946
		ROC	21.2467%	13.8812%	
GBP	Hyperparameters	Hidden size = 128	Hidden size = 128	Hidden size = 256	Hidden size = 256
		Timestep = 141	Timestep = 126	Timestep = 54	Timestep = 33
		Number of Transformer Encoder Layers = 2	Number of Transformer Encoder Layers = 2	Number of LSTM Layers = 3	Number of LSTM Layers = 3
		Number of Heads = 2	Number of Heads = 2	Number of CNN Layers = 1	
		Number of CNN Layers = 1		Kernel Size = 3	
		Kernel Size = 3		Padding = 0	
GBP	Hyperparameters	Padding = 0		Stride = 3	
		Stride = 3			
		Result			
		MSE	0.07869	0.07758	0.07649
		MAE	0.21487	0.22389	0.21760
		ROC	5.8496%	-2.8906%	
GBP	Hyperparameters	Hidden size = 128	Hidden size = 128	Hidden size = 256	Hidden size = 256
		Timestep = 135	Timestep = 9	Timestep = 126	Timestep = 126
		Number of Transformer Encoder Layers = 2	Number of Transformer Encoder Layers = 1	Number of LSTM Layers = 2	Number of LSTM Layers = 3
		Number of Heads = 2	Number of Heads = 4	Number of CNN Layers = 1	
		Number of CNN Layers = 1		Kernel Size = 3	
		Kernel Size = 3		Padding = 0	
GBP	Hyperparameters	Padding = 0		Stride = 3	
		Stride = 3			
		Result			
		MSE	0.06925	0.05121	0.00963
		MAE	0.23517	0.18169	0.21295
		ROC	25.3641%	14.6795%	

sentiment information.

Generally, for the USD/CNY and GBP/CNY datasets, incorporating market sentiment optimizes the models' predictive performance in most cases, and the improvement is significant according to the ROC values. However, for the EUR/CNY dataset, introducing market sentiment seems to be detrimental. The additional market sentiment information almost severely weakens the predictive performance of CNN-LSTM and LSTM models.

We believe this is related to the news text we used. According to our word frequency analysis of the news dataset (Fig. 4), words related to the United States, such as "Dollar", "Federal Reserve", and "USA", appears most frequently. There are also some words related to the United Kingdom, like "pound", but words related to the European Union are less common. Due to the lack of descriptions of the European market in the news dataset, the market sentiment cannot reflect changes in the EU market but can reflect changes in the UK and USA markets, as most news is related to them. For the task of predicting the Euro exchange rate, the market sentiment information related to the USA and UK naturally become noise rather than valuable information in the EUR/CNY dataset. Introducing additional noise evidently weakens the models' performance, leading to increased predictive errors.

In summary, we preliminarily believe that in most cases, introducing market sentiment can improve the performance of prediction models. In other words, market sentiment information helps the model gain a more comprehensive understanding of exchange rate fluctuations, not just based on the exchange rate data itself. This broadens the models' perspective, allowing it to indirectly observe the overall market trend, establish deeper relationships between exchange rate data, and understand patterns of market changes. To gain a more comprehensive understanding of how market sentiment affects model performance, we continue to experiment with changing the timestep in models equipped

with the best hyperparameters, observing the models' sensitivity to timesteps.

Fig. 15 shows the performance of models equipped with the best hyperparameters on datasets with and without market sentiment information, where "*" indicates experiments conducted on datasets with market sentiment information. We observe an intriguing pattern: when the time step length is less than 100, models on datasets with market sentiment information do not consistently outperform those on datasets without it. In some cases, models without market sentiment information perform better than those with it. However, when the timestep exceeds 100, the situation changes. Almost all experimental results of the models suggest that exploiting market sentiment information yields better outcomes. We believe this phenomenon is related to the timeliness of news and the delayed impact of news events.

Ederington and Lee (1993) conducted a study on the impact of regular macroeconomic news on interest rate and foreign exchange futures markets. Their research showed that the release of these news had a significant effect on market volatility, but the time for the impact to take effect could be delayed by several days or even up to a week. Yuan et al. (2020) examined how Trump's tweets affected the stock market, where the delayed effects could only last a few hours. The sentimental information of the news for a certain period of time does not have an immediate impact on the market and needs time to work, so the longer timestep should be more appropriate for the forex forecasting method that is based on news sentimental analysis.

6. Conclusion

Our primary research objective was to develop a novel method for exchange rate prediction, considering the complexity of financial markets and the limitations of using traditional technical methods. Inspired

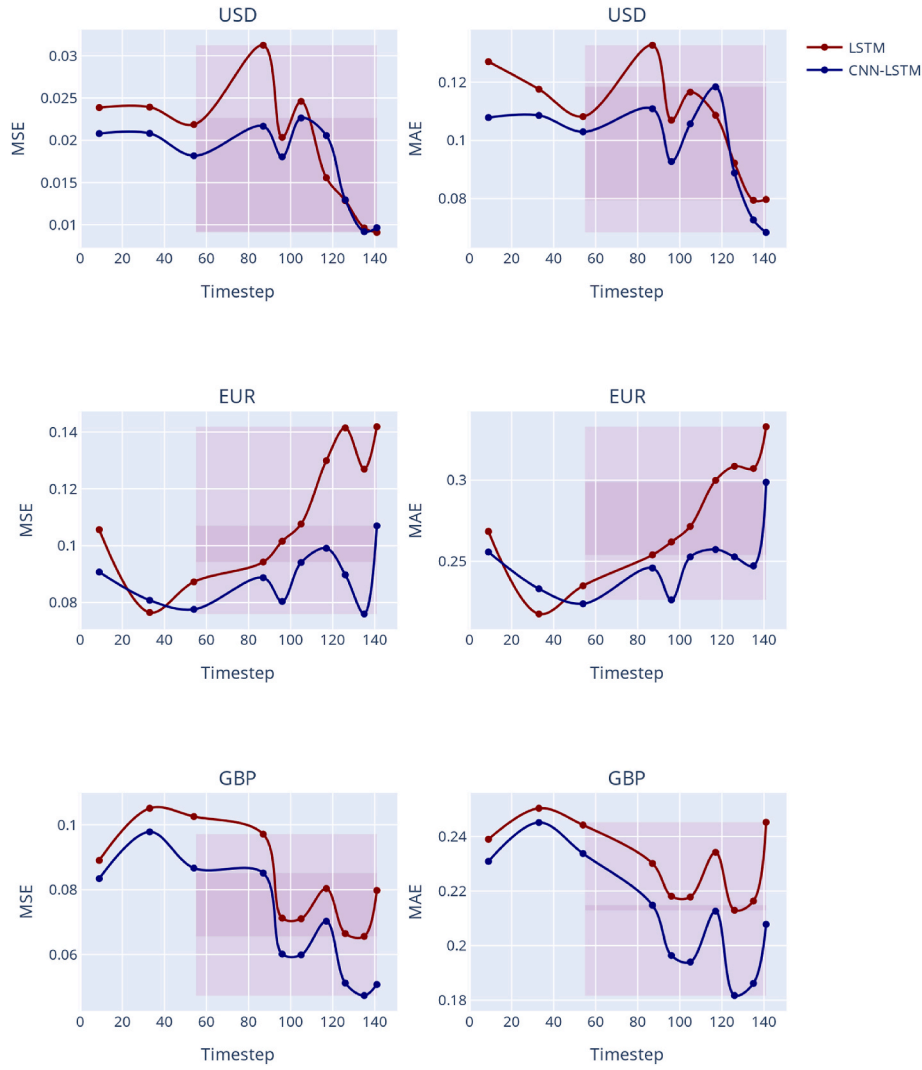


Fig. 14. Model performance at various timestep.

Table 8

Model Performance on Dataset w/o Market Sentiment.

Dataset	Model	CNN-LSTM		LSTM		CNN-Transformer		Transformer	
USD	Market	w	o	w	o	w	o	w	o
	Sentiment								
	MSE	0.00966	0.01416	0.00963	0.01588	0.07101	0.08948	0.12223	0.10112
	MAE	0.06843	0.08847	0.07946	0.09656	0.20757	0.23351	0.26357	0.26083
EUR	Market	w	o	w	o	w	o	w	o
	Sentiment								
	MSE	0.07758	0.02801	0.07648	0.03152	0.07869	0.21843	0.10112	0.13495
	MAE	0.22389	0.13483	0.21760	0.13993	0.21487	0.36180	0.26083	0.27242
GBP	Market	w	o	w	o	w	o	w	o
	Sentiment								
	MSE	0.05121	0.06017	0.06645	0.07367	0.06925	0.32491	0.16259	0.37290
	MAE	0.18169	0.20068	0.21295	0.21842	0.23517	0.45571	0.31509	0.45955
		ROC	9.4628%	2.5043%		48.3948%		31.4350%	

by the theory of CNN's receptive fields, we proposed two models: CNN-LSTM and CNN-Transformer. These models leverage CNN modules to enhance LSTM's learnable timesteps and reduce the complexity of Transformer. Our findings indicate that incorporating CNN modules is an effective way to optimize the performance of LSTM and Transformer. The CNN modules indeed aid LSTM in learning longer time steps.

Generally, CNN-LSTM demonstrates higher predictive accuracy in exchange rate forecasting.

Furthermore, we introduced market sentiment information to enrich the three exchange rate datasets. Our experiments confirmed the effectiveness of this method. However, we also observed that its effectiveness is greatly dependent on the timesteps. Longer timesteps often increase

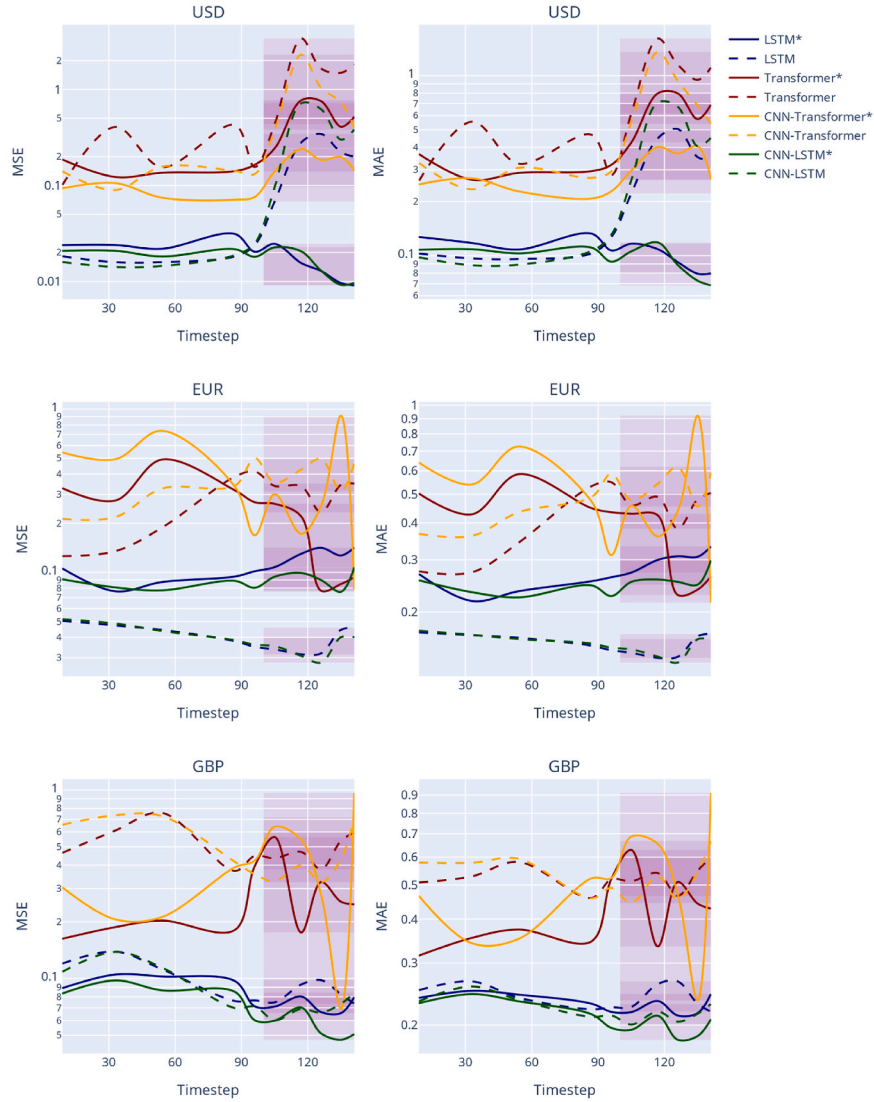


Fig. 15. Model performance at various timestep (w/o market sentiment information).

the likelihood of success for this method. We believe this pattern relates to the timeliness of news and the delayed impact of news events. Also, it's important to note that the content of news text significantly influences the experimental results. In our experiments, the market sentiment extracted was predominantly from news describing US-China and UK-China relations, neglecting European news. This led to a weakening of the models' performance on the EUR/CNY dataset and an enhancement on the USD/CNY and GBP/CNY datasets.

Additionally, our research has practical applications. Improving the accuracy of exchange rate forecasting can assist governments in making informed decisions about national economic restructuring and expansion. It also provides individuals with guidance for cross-border trade and investment opportunities, and offers multinational corporations insights into international business activities like hedging and short-term financing. Particularly in foreign financing, short-term financing decisions consider financing rates and exchange rate fluctuations, underscoring the importance of exchange rate forecasting. Multinational companies must consider currency trends in other countries when making long-term investment decisions, such as establishing subsidiaries in these countries. Thus, forex forecasting is of significant importance.

However, due to hardware and time constraints, we did not experiment with all mainstream exchange rates, selecting only the US dollar, Euro, and British pound. Moreover, our search range for hyperparameters was limited, so we may not have found the strictly best hyperparameters. Additionally, since CNN modules exponentially shorten sequences, longer input sequences would better highlight their function. The longest time step in our experiments was not very long, only 141. Perhaps with much longer time steps, like thousands or tens of thousands, the exponential sequence shortening characteristic of CNN modules would be more evident, but this requires more extensive historical exchange rate data and greater memory capacity. Furthermore, the inherent bias in news information limits prediction accuracy, necessitating more detailed news categorization and more scientific judgment methods.

In summary, our research aims to make a significant contribution by providing a novel method for predicting exchange rates and enhancing accuracy. We believe our findings are of substantial practical importance to various stakeholders and are committed to further refining our methods in future research. Codes and datasets are available at github.com/MaoYuntao/Forex.

CRediT authorship contribution statement

Yuntao Mao: Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ziwei Chen:** Data curation, Writing – review & editing. **Siyuan Liu:** Writing – review & editing. **Yanfeng Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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